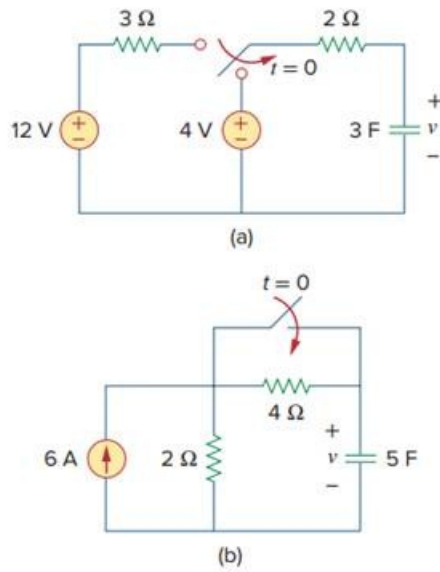


Problem

Find the capacitor voltage for $t < 0$ and $t > 0$ for each of the circuits in Fig. 7.107.

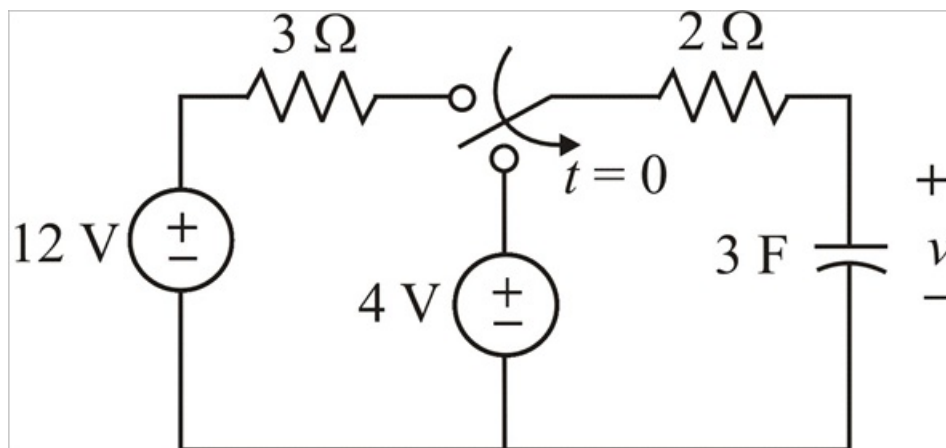
Fig. 7.107:



Step-by-step solution

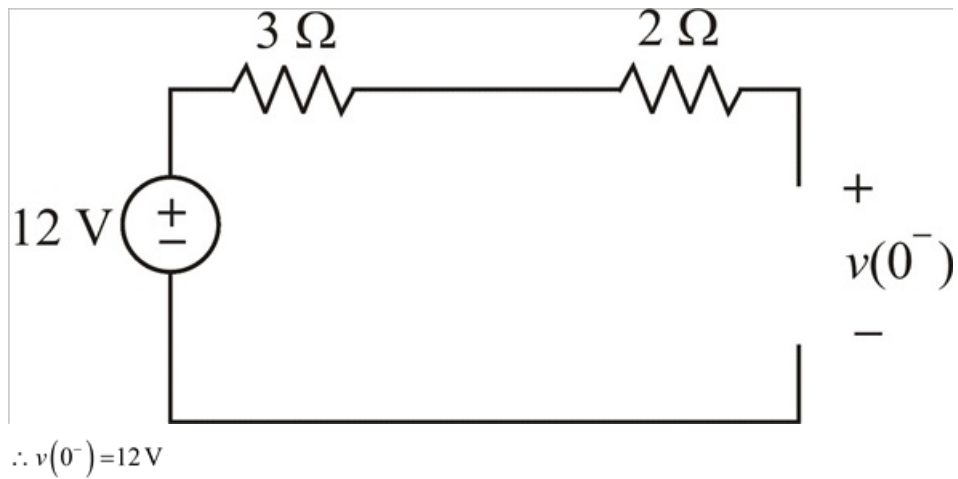
Step 1 of 11

(a) The given circuit is,



Step 2 of 11

For $t < 0$, the switch is at the top position. The capacitor acts like an open circuit to dc, therefore the circuit is



Step 3 of 11

Since the capacitor voltage cannot change instantaneously,

$$v(0^+) = v(0^-)$$

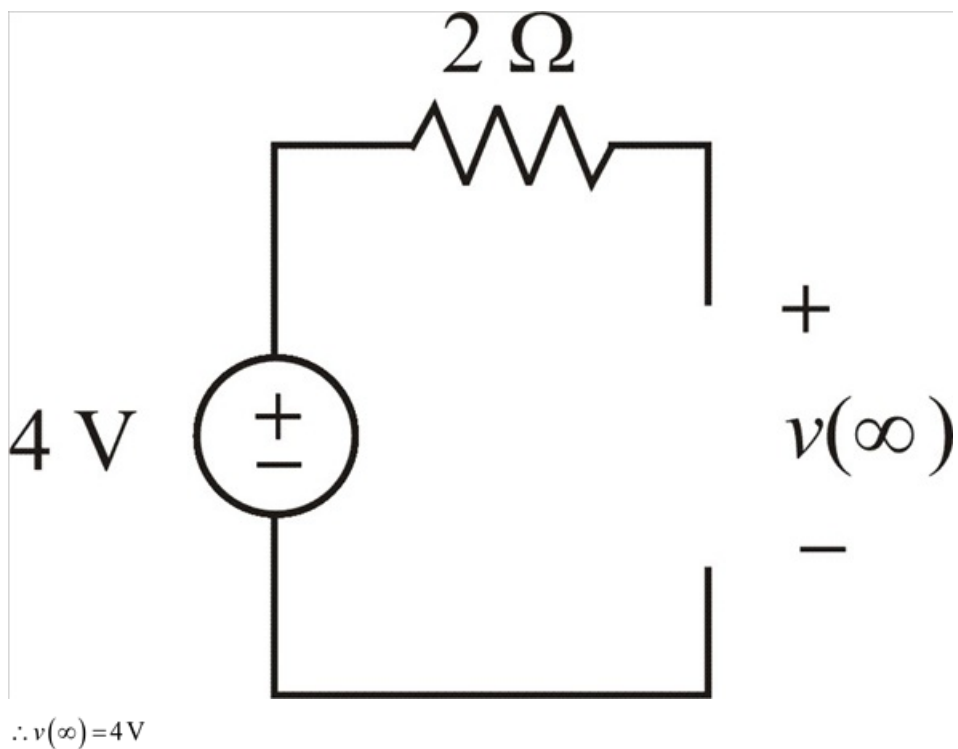
$$\therefore v(0) = v(0^+) = v(0^-)$$

$$v(0) = 12\text{ V}$$

$$\boxed{v(t) = 12\text{ V}} \quad \text{for } t < 0$$

Step 4 of 11

For $t > 0$, the switch is moved to the bottom position. Since the capacitor acts like an open circuit to dc at steady state the circuit is,



Step 5 of 11

The time constant of the circuit is,

$$\tau = RC$$

$$\tau = 2 \times 3$$

$$\tau = 6 \text{ s}$$

Thus,

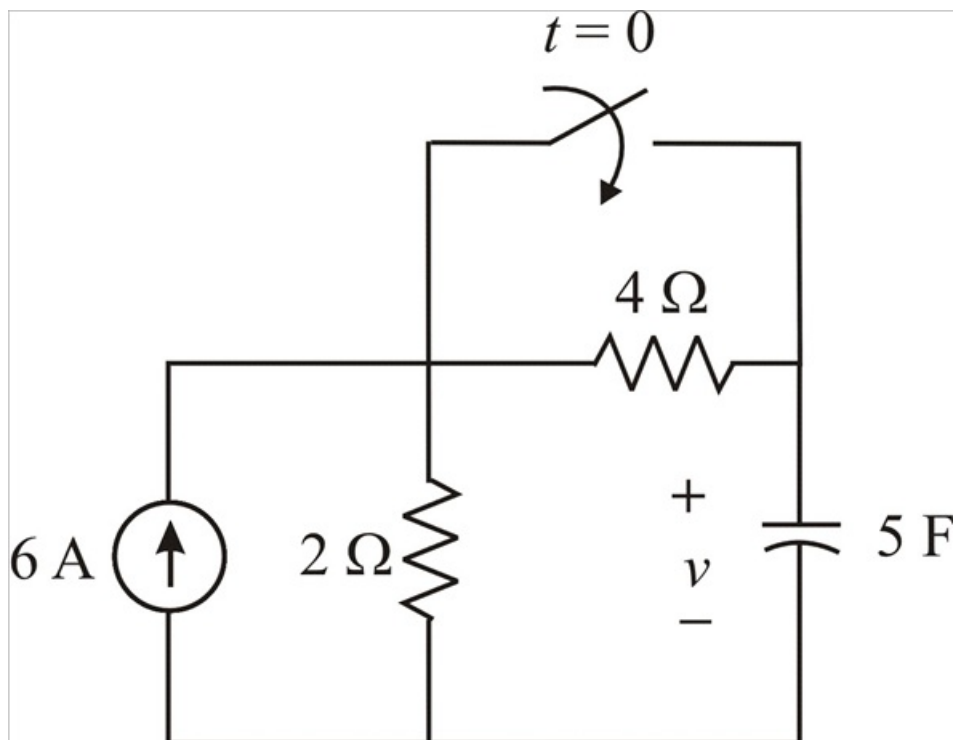
$$v(t) = v(\infty) + [v(0) - v(\infty)]e^{-\frac{t}{\tau}}$$

$$v(t) = 4 + [12 - 4]e^{-\frac{t}{6}}$$

$$\boxed{v(t) = \left(4 + 8e^{-\frac{t}{6}}\right) \text{ V}} \quad \text{for } t > 0$$

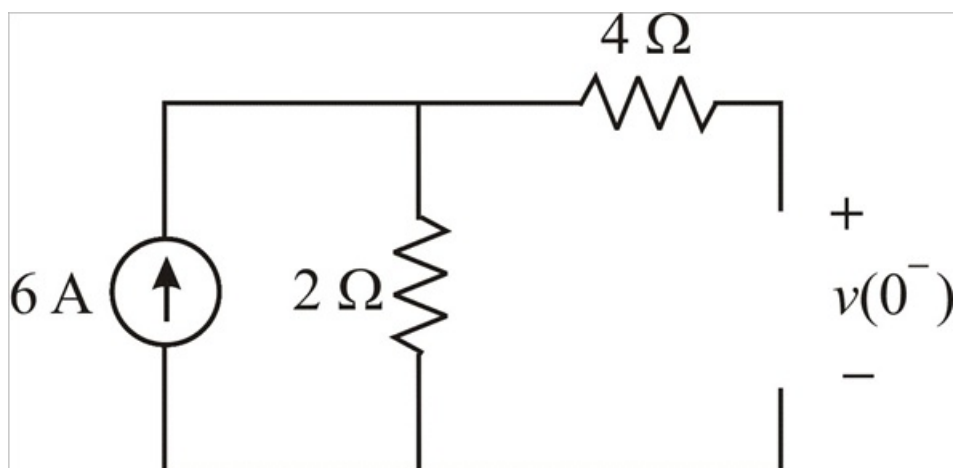
Step 6 of 11

(b) The given circuit is,



Step 7 of 11

For $t < 0$, the switch is in open position. The capacitor acts like an open circuit to dc,



Step 8 of 11

Therefore the voltage across the capacitor is the e as the voltage across the 2Ω resistor,

$$v_{2\Omega} = 2 \times 6$$

$$v_{2\Omega} = 12\text{V}$$

$$v(0^-) = v_{2\Omega}$$

$$v(0^-) = 12\text{V}$$

Since the capacitor voltage cannot change instantaneously,

$$v(0^+) = v(0^-)$$

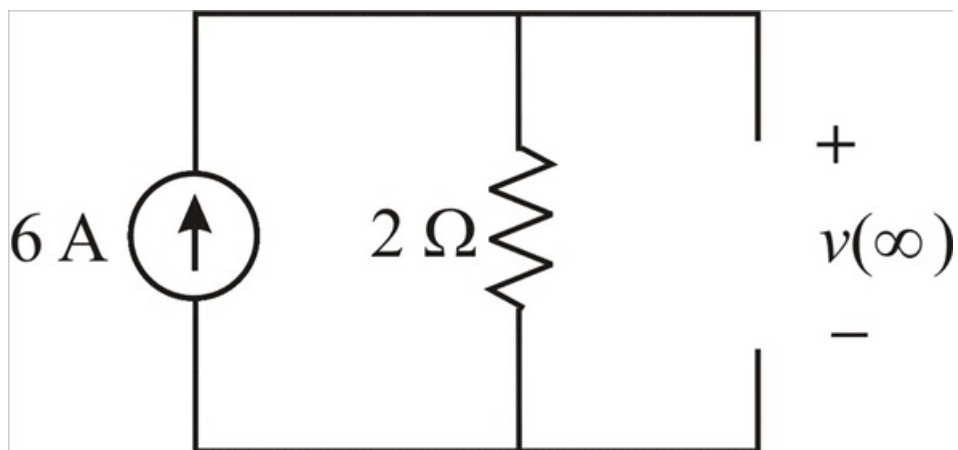
$$\therefore v(0) = v(0^+) = v(0^-)$$

$$v(0) = 12\text{V}$$

$$\boxed{v(t) = 12\text{V}} \quad \text{for } t < 0$$

Step 9 of 11

For $t > 0$, the switch is closed and the 4Ω resistor is removed since it is in parallel with the short circuit, Since the capacitor acts like an open circuit to dc at steady state the circuit is,

**Step 10** of 11

Therefore the voltage across the capacitor is the e as the voltage across the 2Ω resistor,

$$v_{2\Omega} = 2 \times 6$$

$$v_{2\Omega} = 12\text{V}$$

$$v(\infty) = v_{2\Omega}$$

$$v(\infty) = 12\text{V}$$

Step 11 of 11

The time constant of the circuit is,

$$\tau = RC$$

$$\tau = 2 \times 3$$

$$\tau = 6 \text{ s}$$

Thus,

$$v(t) = v(\infty) + [v(0) - v(\infty)]e^{-\frac{t}{\tau}}$$

$$v(t) = 12 + [12 - 12]e^{-\frac{t}{6}}$$

$$\boxed{v(t) = 12 \text{ V}} \quad \text{for } t > 0$$

[Comments \(3\)](#)



Anonymous

The time constant is actually 10, Because capacitance is 5, not 3. $2 \times 5 = 10$



Anonymous

I agree with the last person!



Anonymous

yep agreed